The empirical recurrence for OEIS A239408: recovering a conjecture that is not written down

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Abstract

OEIS A239408 records "Empirical recurrence of order 91", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 100$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 91, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 91 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A239408 is "Number of 4Xn 0..3 arrays with no element equal to one plus the sum of elements to its left or two plus the sum of the elements above it, modulo 4.". It has offset 1 and begins

$$28, 698, 16449, 394509, 9340070, 222457036, 5277728354, 125489197292, \ \ldots$$

The entry states:

> Empirical recurrence of order 91 (see link above)

As of the "Last modified" line on the live entry (Jun 02 2025 09:28:02, revision 7) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 192$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 100$ of the 192, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most* $S' =$ 100.

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 200$ exact terms of the model returns order 91 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{91} c_i\, a(n-i),$$

c_1	22	c_{24}	−179130857208424008	c_{47}	−49610382757399277174999280	c_{70}	1
c_2	431	c_{25}	74102353186443058	c_{48}	−54942891256418435612978322	c_{71}	3
c_3	−9450	c_{26}	−351443244855652057	c_{49}	147238888802027463739111174	c_{72}	−
c_4	−54922	c_{27}	1564274092206169744	c_{50}	−15517518848835713964992998	c_{73}	6
c_5	1525582	c_{28}	−1797156425135436576	c_{51}	−207210372597135166978592658	c_{74}	−
c_6	2255487	c_{29}	2317434985485788262	c_{52}	497286632902514929644270756	c_{75}	−
c_7	−125778478	c_{30}	77920867351266612183	c_{53}	152855676950210492364583644	c_{76}	2
c_8	61100619	c_{31}	−201487361785044893882	c_{54}	−774047937415494779445867700	c_{77}	−
c_9	6007130814	c_{32}	−254789855094215733725	c_{55}	360800291915884861585091496	c_{78}	1
c_{10}	−9738185462	c_{33}	779602618128922787870	c_{56}	−106151797843007180466395728	c_{79}	
c_{11}	−172887652784	c_{34}	−2826464108289257723416	c_{57}	−1125336091203067514140186400	c_{80}	−
c_{12}	429638080449	c_{35}	4209313831294856566560	c_{58}	1153000578495782689723720704	c_{81}	−
c_{13}	2885226679884	c_{36}	22325260528736203469100	c_{59}	−1081094058605984854859930944	c_{82}	
c_{14}	−10110416126581	c_{37}	−31822196988239592159888	c_{60}	−2970664906632233302284252096	c_{83}	
c_{15}	−21748274891538	c_{38}	−4715034575140553909294	c_{61}	6209293486828778036941398720	c_{84}	
c_{16}	134994762384448	c_{39}	45398908118258195884626	c_{62}	4397193670068657847852323392	c_{85}	
c_{17}	−86686993581630	c_{40}	−456646302479015354014243	c_{63}	−7392517690928681107162823424	c_{86}	
c_{18}	−891434670569617	c_{41}	−144692660958129425057500	c_{64}	2663562766070056911980963840	c_{87}	
c_{19}	3152083024154744	c_{42}	1424552887281745427546879	c_{65}	2320201907021334478183931392	c_{88}	
c_{20}	2308314873102	c_{43}	1238442455464395785646020	c_{66}	−9299079577094530585632011776	c_{89}	
c_{21}	−23019347443361874	c_{44}	−1087490243240243298809493	c_{67}	10111224488550137942902293504	c_{90}	
c_{22}	40328582882397771	c_{45}	3595441343010683257126486	c_{68}	−824454299621253634075023360	c_{91}	
c_{23}	39573186407923568	c_{46}	10655594214282817989380134	c_{69}	−16897464869099856677249257472		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 91, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $91 + 4$ terms removed returns order 91 as well, so $\deg q_{\min} = 91 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. ∎

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $91 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 91 that this sequence satisfies — and that family turns out to have exactly one member.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A239408.

[2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.

[3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)

[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.